

Gampa Shreya

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Skills

Programming Languages: Java, Python, JavaScript, HTML/CSS, Bootstrap

Frameworks & Libraries: React.js, Node.js, Express.js

Databases: MongoDB, MySQL

Cloud, Tools & Technologies: AWS, Docker, Git, GitHub, VS Code, Jupyter Notebook

Machine Learning & Deep Learning: CNN, LSTM, GANs, Diffusion Models, Transformers, LLMs, Reinforcement Learning

Core CS Concepts: OOP, Data Structures & Algorithms, Basic System Design Concepts

Experience

AI Research Intern — DrugParadigm

Jul 2025 – Nov 2025

Hyderabad, Telangana, India

On-site

Used **diffusion-based models** for **ADC linker generation**, enabling conditional linker design between antibody and payload .

Optimized **data preprocessing** and feature engineering pipelines, improving model training efficiency and stability.

Designed **training and inference** pipelines to explore and fine-tune different diffusion models for ADC linker generation.

Evaluated generated molecules via RDKit sanitization, novelty analysis, and SMARTS-based pattern filtering to ensure chemical validity.

Education

B.Tech in Computer Science and Engineering, CGPA: 8.57

2023-2027

Keshav Memorial Institute of Technology

Hyderabad

Intermediate

Page Junior College

Hyderabad

Projects

Jewelry Design Pattern Generation | MERN Stack + Deep Learning

- Designed a deep learning model using **GANs** to generate realistic jewelry pattern designs from grayscale sketches.
- **Built a backend** using **Express.js** and **MongoDB** with **RESTful APIs** and **user authentication** for managing user inputs.
- **Developed a frontend** using **React** to upload sketches and visualize AI-generated jewelry designs in real time.

GenAI_Med Diagnosis & Drug Design | Generative AI, Machine Learning

- **Transformer-based** tumor segmentation (**UNETR**) for lung tumor detection from CT scans.
- **Reinforcement learning-based SMILES generation** (stack-augmented RNN) to optimize molecules for target properties.
- Protein structure prediction (ESMFold), visualization, and docking (AutoDock Vina) for ligand–protein interaction analysis.
- Developed a **website** with a React frontend and -Express backend to integrate and visualize model predictions and molecular analysis results.
- Implemented **deployment** of a machine learning model on AWS **EC2**

Achievements

- Solved 400+ DSA problems on LeetCode

Certifications

- Completed Deep Learning with TensorFlow (IBM)
- Achieved Java (Basic) Certification from HackerRank